

MEHMET ALI BULUT

BACKEND SOFTWARE ENGINEER

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SUMMARY

Backend Software Engineer specializing in Java and Spring Boot, with hands-on experience building full-stack applications across internships and personal projects. Skilled in designing scalable REST APIs, layered architectures, and AI-integrated systems — including an LLM-powered financial analysis platform with MCP integration. Comfortable across the full development lifecycle: from system design and unit testing to containerized deployment with Docker.

TECHNICAL SKILLS

Back-End: Java, Spring Boot, Hibernate/JPA, REST & SOAP APIs

Front-End: React.js, JavaScript, Tailwind CSS

DevOps/Tools: Docker, Git, Keycloak, Postman, Redis (Basic), AWS (Basic)

DB: PostgreSQL, MongoDB

OS: Linux

Concepts: Unit Testing (JUnit, Mockito), Agile, Async/Sync Architectures, LangChain4j, MCP Integration

EDUCATION

Computer Engineering – Istanbul Medeniyet University | 2021 – 2026

- Computer Science Core: Advanced Data Structures, Algorithms, Operating Systems
- Systems & Infrastructure: Database Management Systems, Computer Networks
- AI & Data Intelligence: Deep Learning, Data Mining, Pattern Recognition, NLP

PROJECTS

Smart Budgeting App [\[GitHub\]](#)

Java 17, Spring Boot 3.2, React 19, PostgreSQL, Docker, LangChain4j, Apache PDFBox, MCP, AWS

- **Hybrid LLM Router & Cost Optimization:** Designed a smart transaction routing pipeline that classifies PDF bank statement lines locally via regex before invoking GPT-4o-mini, achieving an 87.5% reduction in LLM token consumption with zero accuracy loss.
- **AI Financial Coach (MCP Integration):** Exposed structured financial tools through a Model Context Protocol (MCP) server, enabling an AI coach to query per-user spending data on-demand without embedding entire transaction histories in prompts.
- **Multi-Bank Strategy Pattern & Self-Learning Cache:** Engineered a polymorphic bank parser that auto-detects Turkish bank statement formats and pairs it with a merchant category cache, eliminating repeated LLM calls for known merchants.
- **Security & Reliability:** Implemented JWT authentication with HttpOnly cookie-based refresh token rotation, per-IP rate limiting via Bucket4j, and Flyway database migrations within a fully containerized Docker environment.

Brokerage Firm App [\[GitHub\]](#)

Java, Spring Boot, React, Docker, PostgreSQL, JUnit/Mockito

- **Brokerage System Engineering:** Engineered an end-to-end full-stack architecture for a stock trading platform using Java, Spring Boot, and React, focusing on secure transaction handling and system scalability.
- **System Reliability & Persistence:** Architected a layered architecture with PostgreSQL integration and Docker containerization, ensuring high reliability through comprehensive unit tests with JUnit and Mockito.

Nearby Places Finder [\[GitHub\]](#)

Java, Spring Boot, React, Spring WebFlux

- **Reactive System Architecture:** Engineered an end-to-end web application utilizing Spring WebFlux for non-blocking, reactive API calls to ensure high-performance coordinate-based data retrieval.
- **Geospatial Integration:** Integrated Leaflet.js for interactive map functionality, enabling seamless visualization of location-based data through a modern React frontend.

EXPERIENCE

Long-term Intern — [Finartz](#) | Istanbul, Turkey | Nov 2025 – Jan 2026

- **Java, Spring Boot & React:** Engineered a full-stack brokerage platform utilizing a layered architecture to maximize system autonomy and scalability.
- **PostgreSQL & Docker:** Integrated robust relational data management and containerized the application environment to ensure seamless consistency across development and deployment stages.
- **Quality Assurance & Agile:** Guaranteed high system reliability and clean code standards by architecting comprehensive unit tests (JUnit/Mockito) within an Agile and OOP-driven framework.

Intern — [Codexist](#) | Istanbul, Turkey | July 2023 – Sept 2023

- **Java & Spring Ecosystem:** Developed a banking application following enterprise architecture standards, focusing on RESTful API design and modern dependency management.
- **Structured Information Repository (JDBC):** Engineered a backend system to manage and persist structured data, strengthening core skills in database operations, data integrity, and CRUD processes.
- **Software Development Practices:** Applied industry-standard workflows to ensure high-quality delivery and alignment with modern software engineering principles.

ADDITIONAL INFORMATION

Languages: English (B1+)

Certifications:

TUBITAK BILGEM YTE Bootcamp: Java (2023), Spring Boot (2023)

Aygaz / Global AI Hub: Image Processing (2025)

References: Available upon request